

Control Variates: Variance Reduction and an Example

Optimal variance reduction

With $\beta^* = \text{Cov}(h(X), c(X)) / \text{Var}(c(X))$,

$$\text{Var}[h_{\beta^*}(X)] = \text{Var}[h(X)] (1 - \rho^2),$$

where $\rho = \frac{\text{Cov}(h(X), c(X))}{\sqrt{\text{Var}(h(X)) \text{Var}(c(X))}}$. Derivation:

$$\text{Var}(h_{\beta}) = \text{Var}(h - \beta(c - \mu_c)) = \text{Var}(h) + \beta^2 \text{Var}(c) - 2\beta \text{Cov}(h, c).$$

Plugging in β^* gives

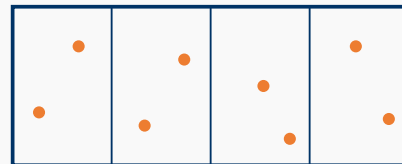
$$\text{Var}(h_{\beta^*}) = \text{Var}(h) - \frac{\text{Cov}(h, c)^2}{\text{Var}(c)} = \text{Var}(h) (1 - \rho^2).$$

Since $0 \leq \rho^2 \leq 1$, we have $\text{Var}(h_{\beta^*}) \leq \text{Var}(h)$, with **strict** reduction when $|\rho| > 0$.

$\int h(x) dx$
 $\rightarrow c(x) \nmid h(x)$
 $\approx$ correlated
 $E(c(x_1)) = \mu$

Stratified Sampling: Idea and Algorithm

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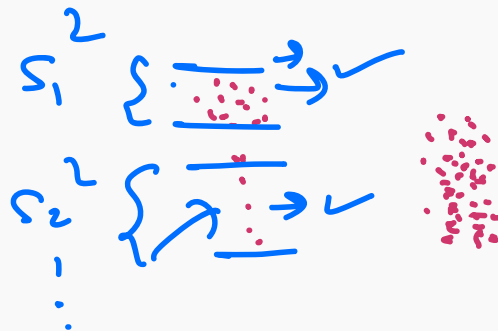
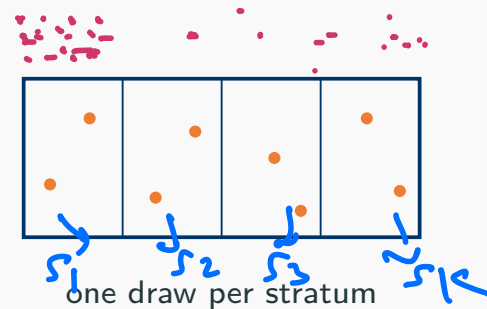


one draw per stratum

$$\begin{aligned}
 & \int g(x) dx \\
 &= \int \frac{g(x)}{f(x)} f(x) dx = E \left[\frac{g(x)}{f(x)} \right] \\
 \hat{\theta}_n &= \frac{1}{n} \sum_{i=1}^n \frac{g(x_i)}{f(x_i)} \\
 SE(\hat{\theta}_n) &= \frac{\sigma}{\sqrt{n}} \rightarrow
 \end{aligned}$$

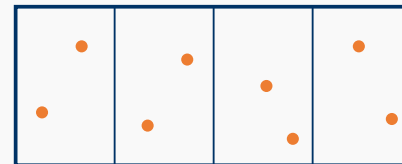
Stratified Sampling: Idea and Algorithm

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- **Idea:** partition the domain into K strata of known probability p_k , draw a fixed number from *each* stratum, and combine.



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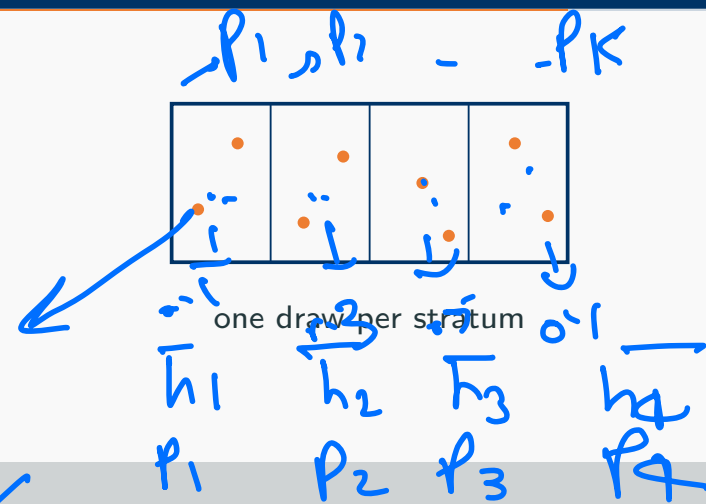
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Algorithm: Stratified Sampling

1. Partition $\mathcal{X}$ into K disjoint strata with known probabilities $p_1, \dots, p_K$ ($\sum p_k = 1$).
2. In stratum k , draw n_k points from f restricted to that stratum; compute the local mean $\bar{h}_k$.

3. Return $\hat{\theta}_n^{\text{ST}} = \sum_{k=1}^K p_k \bar{h}_k$.

Handwritten notes and formulas:

$$\hat{\theta}^{\text{mc}} = \frac{1}{n} \sum_{i=1}^n h(x_i)$$
$$\hat{\theta}^{\text{ST}} = \sum_{k=1}^K p_k \bar{h}_k$$

Stratified Sampling: Why It Helps

Variance identity

Writing $\sigma_k^2 = \text{Var}(h(X) \mid X \in \text{stratum } k)$ and proportional allocation $n_k = n p_k$,

$$\text{Var}[\hat{\theta}_n^{\text{ST}}] = \frac{1}{n} \sum_{k=1}^K p_k \sigma_k^2 \leq \frac{1}{n} \text{Var}[h(X)] = \text{Var}[\hat{\theta}_n^{\text{crude}}].$$

Stratification **can only help** (or leave variance unchanged) because it removes the “between-stratum” component of variance.

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- **Climate application:** when estimating expected regional temperature anomaly under uncertain climate sensitivity, stratifying draws by decade or emissions scenario guarantees every scenario is represented, rather than leaving it to chance.
- Works best when the domain can be naturally partitioned into strata with genuinely different behaviour of h .

Which Technique When?

Technique	Needs	Best suited for
✓ Antithetic variates	Monotonic h	Cheap, quick win; path simulation
✓ Control variates	Correlated $c(X)$ with known $E[c(X)]$	A related, easier problem is solvable
✓ Stratified sampling	A natural partition of the domain	Domain has distinct regimes/regions
✓ Importance sampling	A good proposal g	Rare events heavy tails, un-normalized f

$$E f(\mu) = E g \left(\frac{f(x)}{g(x)} h(\mu) \right)$$

→ $x_1 \dots x_n \sim f$

IS
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These are not mutually exclusive

In practice, serious Monte Carlo pipelines (option pricing engines, rendering software, Bayesian samplers) routinely **combine** several of these: e.g. importance sampling *plus* antithetic pairs *plus* a control variate, all at once.

Outline

Motivation: The Integrals We Cannot Solve

The Monte Carlo Principle

Law of Large Numbers and Monte Carlo Error

Crude Monte Carlo in Practice

Importance Sampling

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Practical Monte Carlo Diagnostics

Trace Plots and Running Means

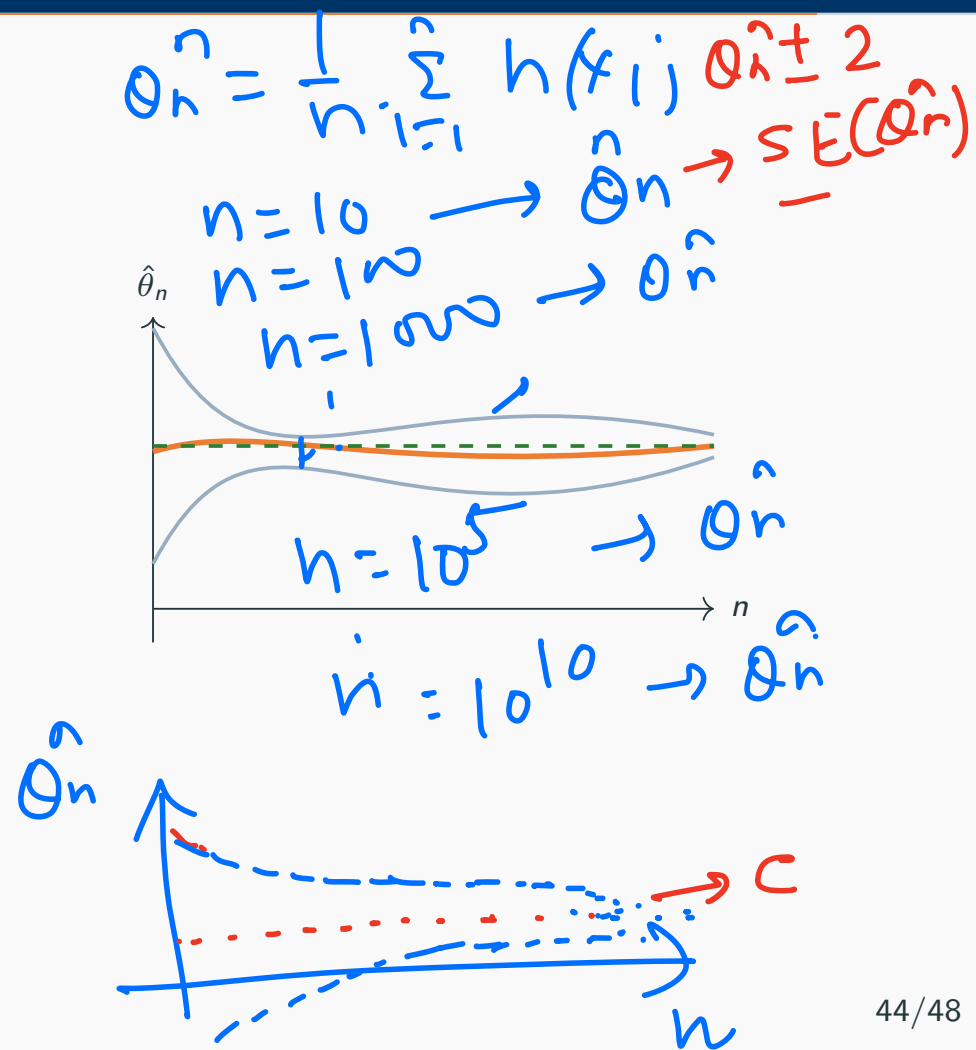
Effective Sample Size for Importance Sampling

Honest Reporting, Revisited

Unit Summary

Diagnostic 1: The Running-Mean Trace Plot

- Plot $\hat{\theta}_k = \frac{1}{k} \sum_{i=1}^k h(X_i)$ against k , alongside the approximate ± 2 SE band.
- A well-behaved estimator: the trace should settle down and the band should visibly narrow like $1/\sqrt{k}$.
- Wild, non-shrinking swings suggest infinite (or very large) variance – a red flag, especially common with poorly chosen importance sampling proposals.



Diagnostic 2: Effective Sample Size (ESS)

- In importance sampling, some draws matter far more than others – a highly **uneven** weight distribution means we effectively have far fewer than n “useful” draws.

$$x_1 \dots x_n \sim f \quad \rightarrow \quad \begin{aligned} w_1 &= \frac{f(x_1)}{g(x_1)} \\ &\vdots \\ w_n &= \frac{f(x_n)}{g(x_n)} \end{aligned}$$

↕

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- A widely used diagnostic (Kong, 1992):

$$\text{ESS} = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2}, \Rightarrow 1 \leq \text{ESS} \leq n.$$

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Rule of thumb

If ESS is only a small fraction of n (e.g. $\text{ESS}/n < 0.1$), reconsider the proposal g – it likely has tails that are too light relative to the target.

Putting the Diagnostics Together

A responsible Monte Carlo workflow

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Why this matters

A single Monte Carlo number with no diagnostics is not a scientific result – it is a guess dressed up in decimal places. These checks are what separate a trustworthy simulation study (in finance, genomics, graphics, or anywhere else) from a silently broken one.

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- Any hard integral $\theta = \int hf \, dx$ can be rewritten as an expectation $E_f[h(X)]$ and estimated by a sample average – **dimension-free** error at rate $O(1/\sqrt{n})$.

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- **Variance reduction** techniques squeeze more precision out of the same computational budget.
- Always **diagnose** the result: standard error, trace plots, effective sample size.